

CUI Peng

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Educational Background

Hong Kong University of Science and Technology September 2025 – October 2026
Master of Science in Artificial Intelligence
Southern University of Science and Technology September 2021 – June 2025
Bachelor of Science in Data Science and Big Data Technology
GPA: 3.59/4.0 & Weighted Average Score: 86.92/100
TOEFL: 90/120

Papers

- **Bayesian Gated Non-Negative Contrastive Learning.** Peng Cui*, Jiahao Zhang*, Lijie Hu. *International Conference on Machine Learning (ICML), 2026.* (*: Equal Contribution)
- **Complementary Concept Bottleneck Models.** Jiahao Zhang*, Peng Cui*, Hongbin Lin, Xinyue Xu, Lijie Hu. *Working Paper.* (*: Equal Contribution)

Personal Projects

- **LLM Fine-tuning Data Selection** - A systematic framework for implementing and comparing different data selection algorithms.
- **DiffFake** - Difficulty aware data selection in deep fake image detection.
- **Modular RAG** - A modular, scalable retrieval-augmented generation system based on the Chroma vector database. Users can select appropriate methods at different stages of RAG.
- **Telegram Academic Bot** - A AI assistant powered by a large language model that provides automatic crawling, aggregation, personalized recommendations and intelligent Q&A.

Research Experience

Guangming Institute of Advanced Studies, SUSTech July 2025 – August 2025
Visiting Research Student, Supervisor: Prof. Hongxin Wei

- Mainly explores fine-tuning data selection and model merging.

MBZUAI September 2025 – Ongoing
Visiting Research Student, Supervisor: Prof. Lijie Hu

- Mainly explores explainable AI and large language model unlearning.

HKUST SmartX Lab March 2026 – Ongoing
Research Intern, Supervisor: Prof. Hao Chen

- MRI Medical Image Analysis

Industry Experience

Beijing Shangqi Digital Technology Co., Ltd.

September 2024 – November 2024

Big Data Development Intern

- Responsible for data collection and pre-processing of big data platforms.
- Automated extraction and synthesis of required data using Python scripts.

Shenzhen Mengwang Technology Development Co., Ltd.

April 2025 – June 2025

AI Algorithm Intern

- Data synthesis for e-commerce shopping guide, including general dialogue data and tool-use data.

Project Experience

A Data Collector (ZL2024 2 0757992.1)

Granted in October 2024

Technical Leader

- Developed a new model of the data collector, which solved the problem that the connection head of the existing data collector is easily lost during use due to vibration of internal components and the cooling system, resulting in inaccurate data collection.
- The product can be applied to barcode scanning devices, RFID technology, data terminal integration, and offline terminal computer equipment.

Extracurricular Activities

- Head of Development Department, Students' Association Union – raised more than RMB 30,000 of funds for school events, e.g. Campus Singing Competition, or Association Recruitment Event.